25900773 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCCHE100 - FUNDAMENTALS OF CHEMISTRY-I

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

1. Which of the following is not a spectral series in the line spectrum of hydrogen? [K] / [CO1]

   ☐ Lyman ☐ Balmer

   ☐ Hund's ☐ Paschen

2. The bond angle associated with sp2 hybridisation is [K] / [CO2]

   ☐ 120° ☐ 90°

   ☐ 180° ☐ 109.5°

3. What type of bond is present in a water molecule [U] / [CO5]

   ☐ Ionic bond ☐ Covalent bond

   ☐ Coordinate bond ☐ Metallic bond

4. What is hyperconjugation? [U] / [CO3, CO4]

   ☐ Delocalization of π-electrons ☐ Delocalization of σ-electrons

   ☐ Delocalization of lone pair electrons ☐ Delocalization of d-electrons

5. According to de Broglie’s hypothesis, the wavelength of electron is: [A] / [CO1]

☐ Directly proportional to velocity of particle ☐ Directly proportional to square of velocity of particle

☐ Inversely proportional to velocity of particle ☐ Inversely proportional to square of velocity of particle

6. The order of acidity from strongest to weakest among these acids is: [U] / [CO3, CO4]

☐ Acetic acid > Chloroacetic acid > Fluoroacetic acid ☐ Fluoroacetic acid > Chloroacetic acid > Acetic acid

☐ Chloroacetic acid > Acetic acid > Fluoroacetic acid ☐ All have the same acidity

7. What is the valency of sodium? [U] / [CO5]

☐ 1 ☐ 2

☐ 3 ☐ 4

8. Which of the following quantum numbers governs the spatial orientation of an atomic orbital? [A] / [CO1]

☐ Spin quantum number ☐ Magnetic quantum number

☐ Azimuthal quantum number ☐ Principal quantum number

9. C-O, C-N, and C-H bonds are considered polar in organic mlecules. Report the reason. [U] / [CO3, CO4]

☐ Due to high electronegativity of oxygen, nitrogen and chlorine ☐ Due to absence of lone pairs on carbon

☐ Due to the lack of hybridisation ☐ Due to the covalent radius of carbon

10. Atomic radius decreases across a period due to: [U] / [CO5]

☐ Decreasing effective nuclear charge. ☐ Increasing effective nuclear charge.

☐ Increasing number of electron shells. ☐ Increasing shielding effect.

[10x1 = 10]

Part B

Short Answer Type Questions

Answer any **4** Questions

Each Question carries **3** marks

11. State Pauli's Exclusion Principle and provide an example [U] / [CO1]

12. How does electronegativity vary across a period and down a group in the periodic table? Provide reasons for these trends. [U] / [CO5]

13. Discuss the significance of catenation in organic chemistry [U] / [CO2]

14. What is effective nuclear charge (Zeff)? [U] / [CO5]

15. State Hund's rule of maximum multiplicity and provide an example [U] / [CO1]

16. Write short notes on carbanions. [K] / [CO3, CO4]

[4x3 = 12]

Part C

Short Essay Type Questions

Answer any **4** Questions

Each Question carries **7** marks

17. Explain the limitations of Bohr theory. [U] / [CO1]

18. How does hyperconjugation differ from the inductive effect in stabilizing carbocations? Provide examples to explain their roles. [An] / [CO3, CO4]

19. Explain the differences in properties between d-block elements and those of s-block and p-block elements. [U] / [CO5]

20. What is an orbital? Sketch the shapes of s, p and d orbitals. [U] / [CO1]

21. Explain the delocalization of electrons, the concept of contributing resonance structures, and the impact of resonance on the stability of a molecule. [U] / [CO3, CO4]

22. Explain the reasons behind the anomalous properties of elements like nitrogen, fluorine and oxygen within their respective groups. [A] / [CO5]

[4x7 = 28]